

Artificial Intelligence in Modern Lymphedema Surgery: From Diagnosis to Precision Navigation

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Causes

- Surgery or cancer-related surgery
- Radiation therapy
- Severe trauma
- Tissue infection
- Congenital causes of unknown origin

Clinical Manifestations

- Sensation of heaviness, swelling, and occasional pain
- Most commonly affects one side of the limb
- Increased susceptibility to cellulitis
- Advanced lymphedema may lead to skin changes:
 - Hyperpigmentation
 - Verrucous proliferation
 - Hyperkeratosis
- In severe cases, elephantiasis may develop



Possible Signs and Symptoms of Lymphedema

- **Swelling** and a sensation of **heaviness** in the affected limb. Pressing the skin with a finger may leave an indentation (pitting edema).
- The skin may become rough, **thickened**, firm, or feel tight.
- Reduced **joint mobility** and difficulty moving the limb.
- Decreased **skin integrity**, making the skin more fragile and prone to breakdown.

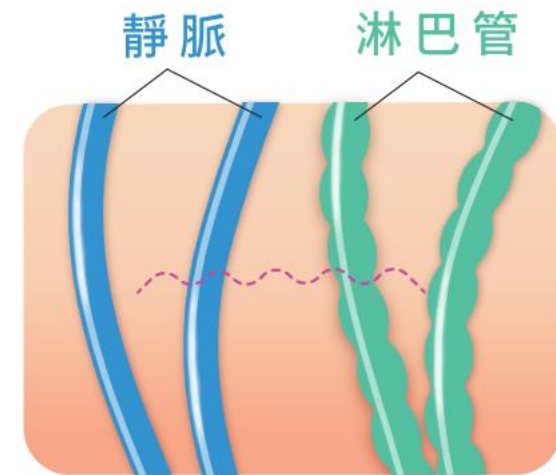
Treatment of Lymphedema

Conservative Management

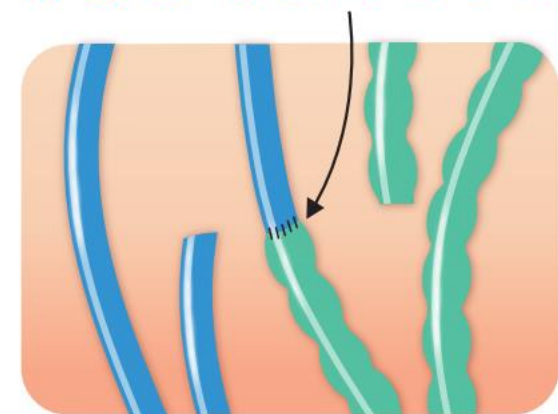
- Limb elevation and exercise
- Compression bandaging
- Manual lymphatic drainage (MLD) / limb massage
- Pneumatic compression therapy
- Compression garment therapy

Physiologic Surgical Treatment

- **Lymphaticovenous anastomosis (LVA)**
- Vascularized lymph node transfer (VLNT)



淋巴管-靜脈吻合手術



Outcomes After Lymphaticovenous Anastomosis (LVA)

- Improvement in limb circumference

- Reduction in limb pain and discomfort

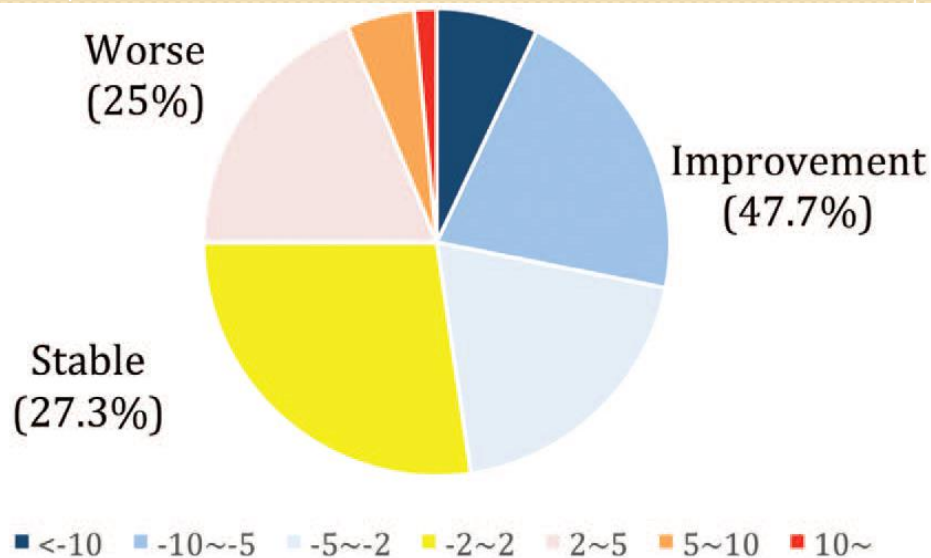
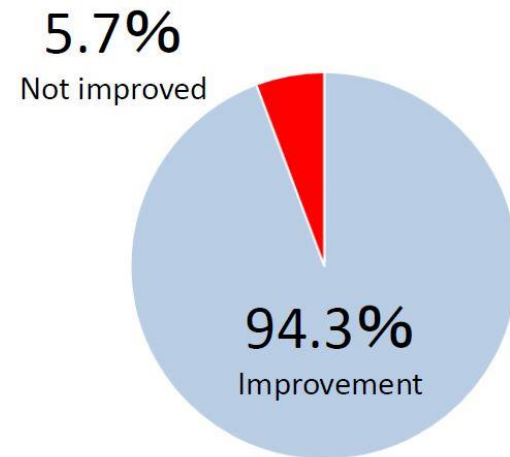


Fig. 1. The preoperative and postoperative distribution of limb circumference change rate. The change rate was calculated using the following equation: $\text{Change rate (percent)} = (\sum \text{Postoperative circumference}) - (\sum \text{Preoperative circumference}) / \sum \text{Preoperative circumference} \times 100$. Of the subjects, 47.7 percent showed improvement in limb circumference postoperatively, 27.3 percent (–2 to 2 percent) showed no change, and 25.0 percent showed exacerbation.

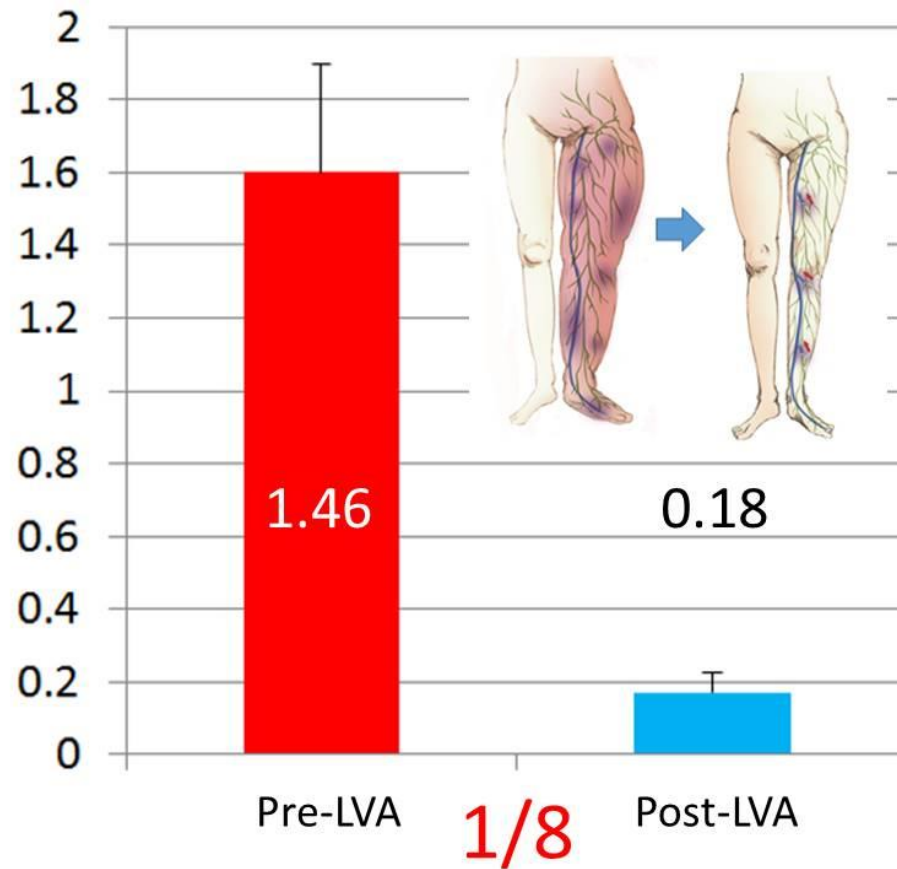
Chronic Pain & Discomfort



'AS scale' is Used by Randomly-Selected Health Professionals in our hospital.

Mihara, Hara et al. PRS 2016 July

Frequency of Cellulitis

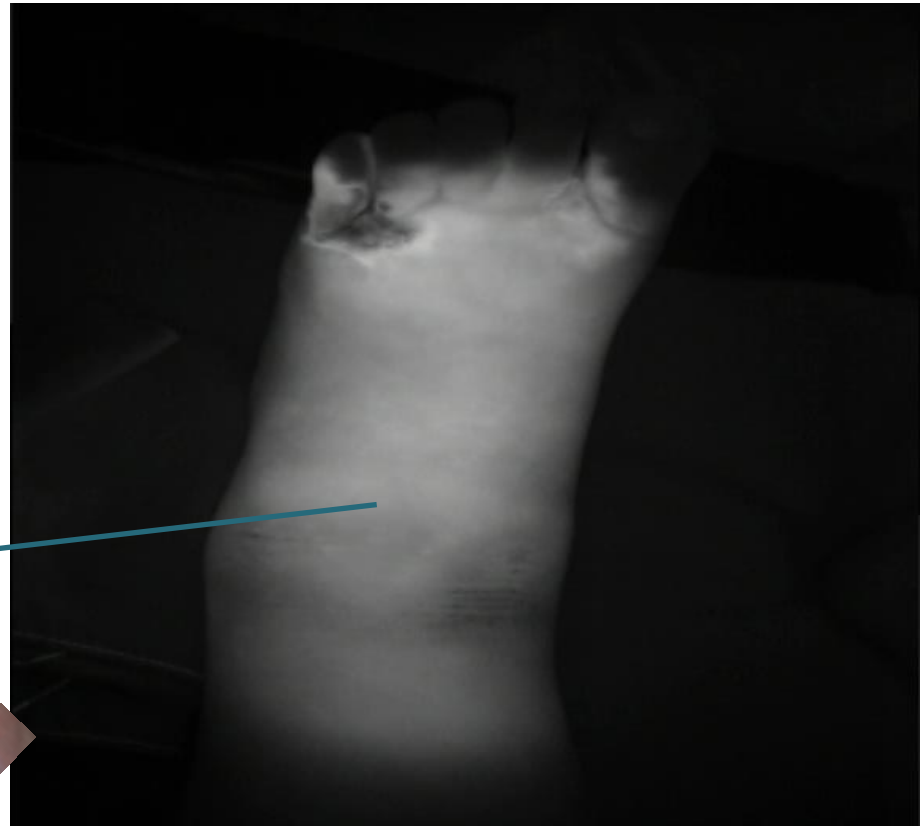
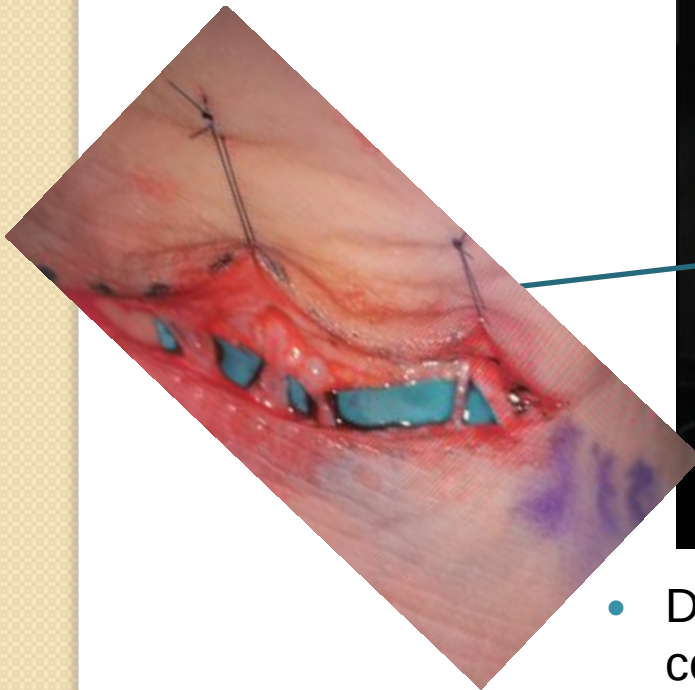


Factors Associated with Good Prognosis

- Presence of multiple functional lymphatic channels
- Compliance with postoperative compression therapy
- Body weight control and BMI reduction

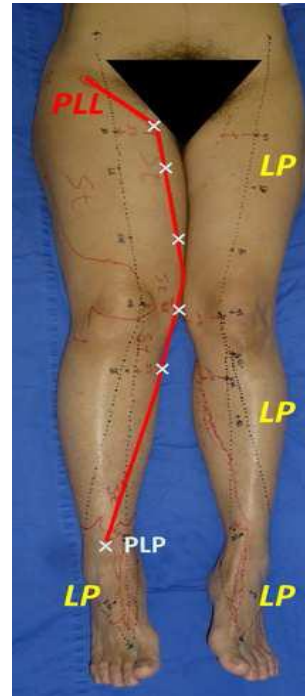
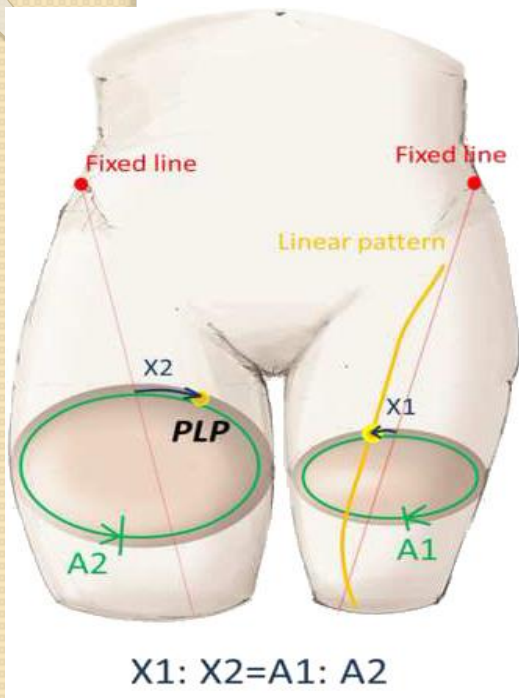
How to Identify Ideal Lymphatic Vessels?

Moderate to
Severe
Lymphedema



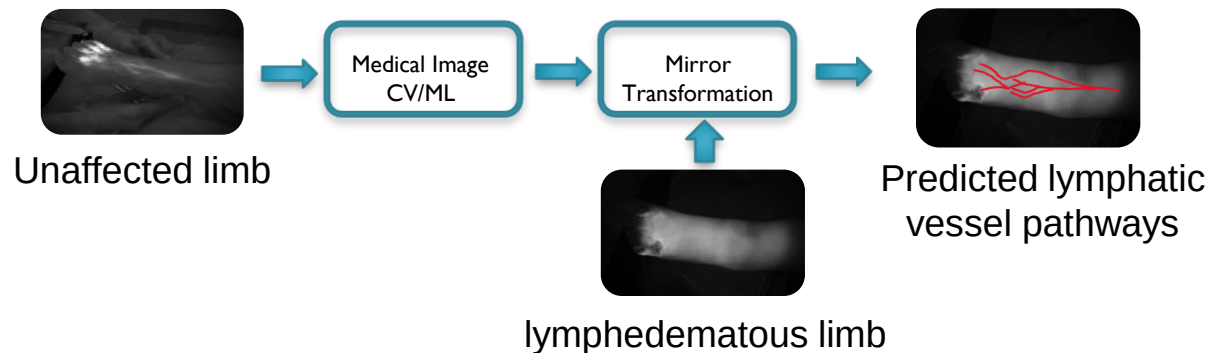
- Diffuse “stardust” or dermal backflow patterns are commonly observed on indocyanine green (ICG) lymphography

Mirror Mapping

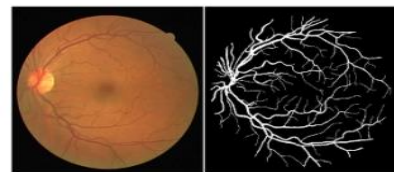


Research Methods / Workflow

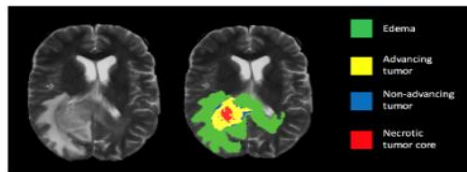
- **Automatic Detection of Lymphatic Vessel Pathways**
 - unaffected limb are mirrored and transformed to predict the lymphatic vessel course in the lymphedematous limb.



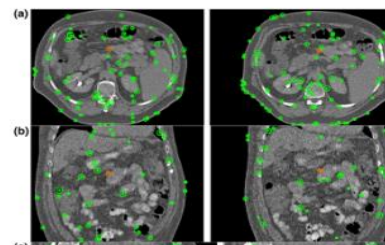
- **Computer Vision and Machine Learning**
 - Edge detection 、 Image segmentation 、 Keypoint detection 、 Structure-from-motion reconstruction



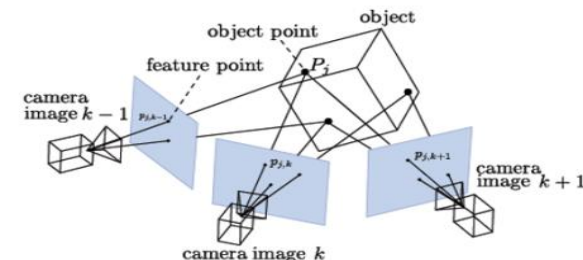
邊界偵測



影像分割



關鍵點偵測



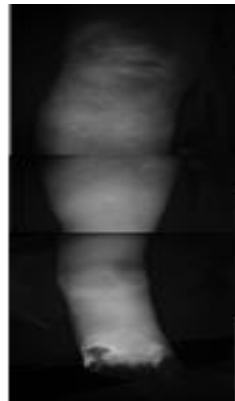
運動恢復結構

Methods

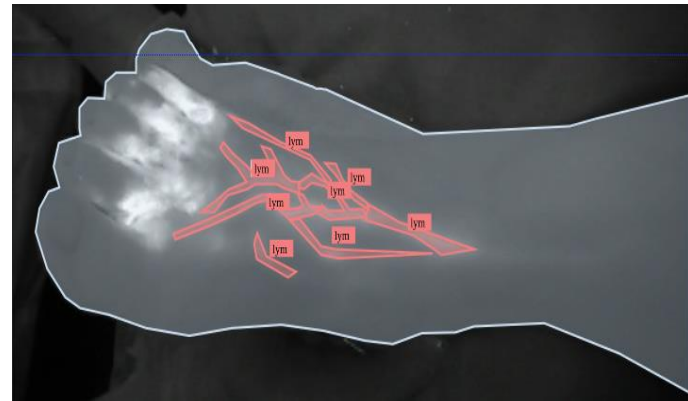
- Development of supervised learning algorithms for lymphatic vessel analysis
- Requirement for large-scale, high-quality annotated datasets
- Development of semi-supervised learning algorithms
 - To overcome the limitation of insufficient labeled data
 - To maintain the recognition performance of supervised learning models
- Continuous collection and annotation of lymphatic pathways in patients with lymphedema

Major Milestones

- Familiarization with near-infrared imaging datasets
- Development of an annotation tool
- Preliminary testing of computer vision models
- Completion of dataset annotation
- Training of computer vision–based segmentation models
- Initial testing of lymphatic vessel pathway detection in unaffected limbs
- Initial testing of lymphatic vessel pathway detection in patients with early-stage extremity lymphedema



Proposal of a novel data acquisition strategy



Development of an annotation platform/tool

Preliminary Computer Vision Model Testing

Origin

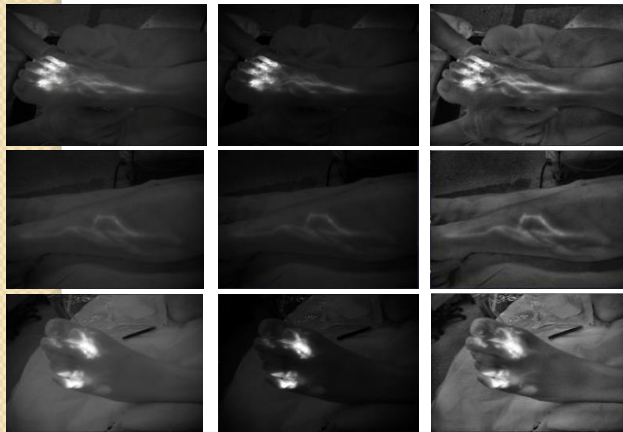
Gamma

CLAHE

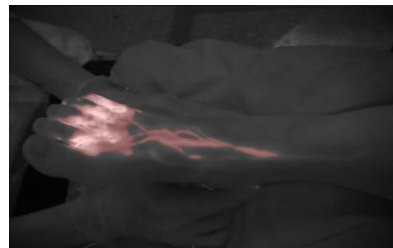
Normal

Normal

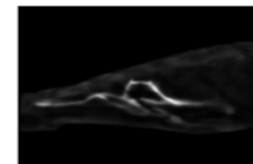
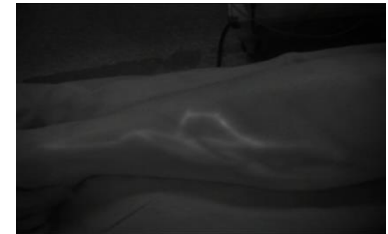
Abnormal



Data Preprocessing



Preliminary Testing
(Successful)



Preliminary Testing (Failed) →
Manual Adjustment Required



ICG image of
unaffected leg



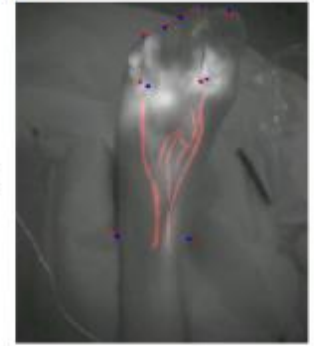
Segmentation



Keypoint detection



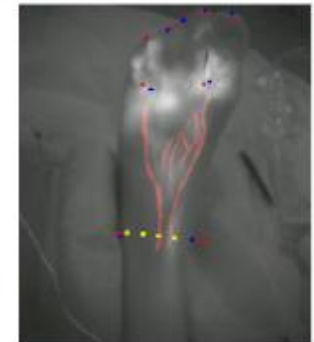
unaffected



Lymphoedema



Recommended
incisional
position
(推薦下刀位置)



淋巴管偵測系統(Lymphatic Vessel Detection)



One of the currently medical solutions is base on the symmetry of the human body.

X: The length of a circumference on normal limb. X': The length of a circumference on edema limb.

Y: The length of a line between the landmark to the lymphatic vessel.

Y': The length of a line between the landmark to the predictive lymphatic vessel.

$$X : X' = Y : Y'$$

$$\Rightarrow Y' = X'Y / X$$



Load Images

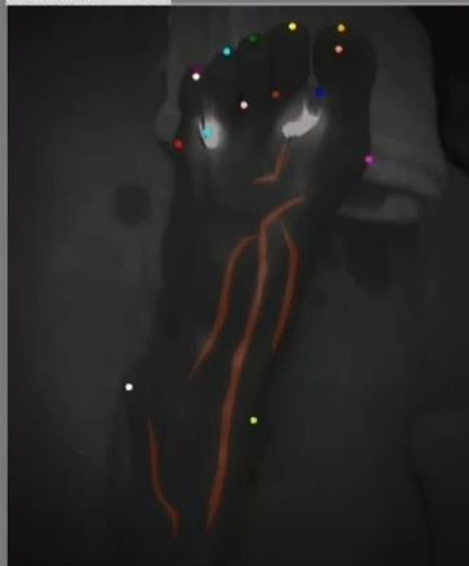
/home/nash/Desktop/Data/tt Load

Guide:

ICG Procrustes distance is 0.013888377331810475

RGB Procrustes distance is 0.013258306070813908

Normal ICG



Edema ICG



Load Images

/home/nash/Desktop/Data/Light/Test/112092702 Load

Guide:

ICG Procrustes distance is 0.009983385135135303

RGB Procrustes distance is 0.013701797790599546

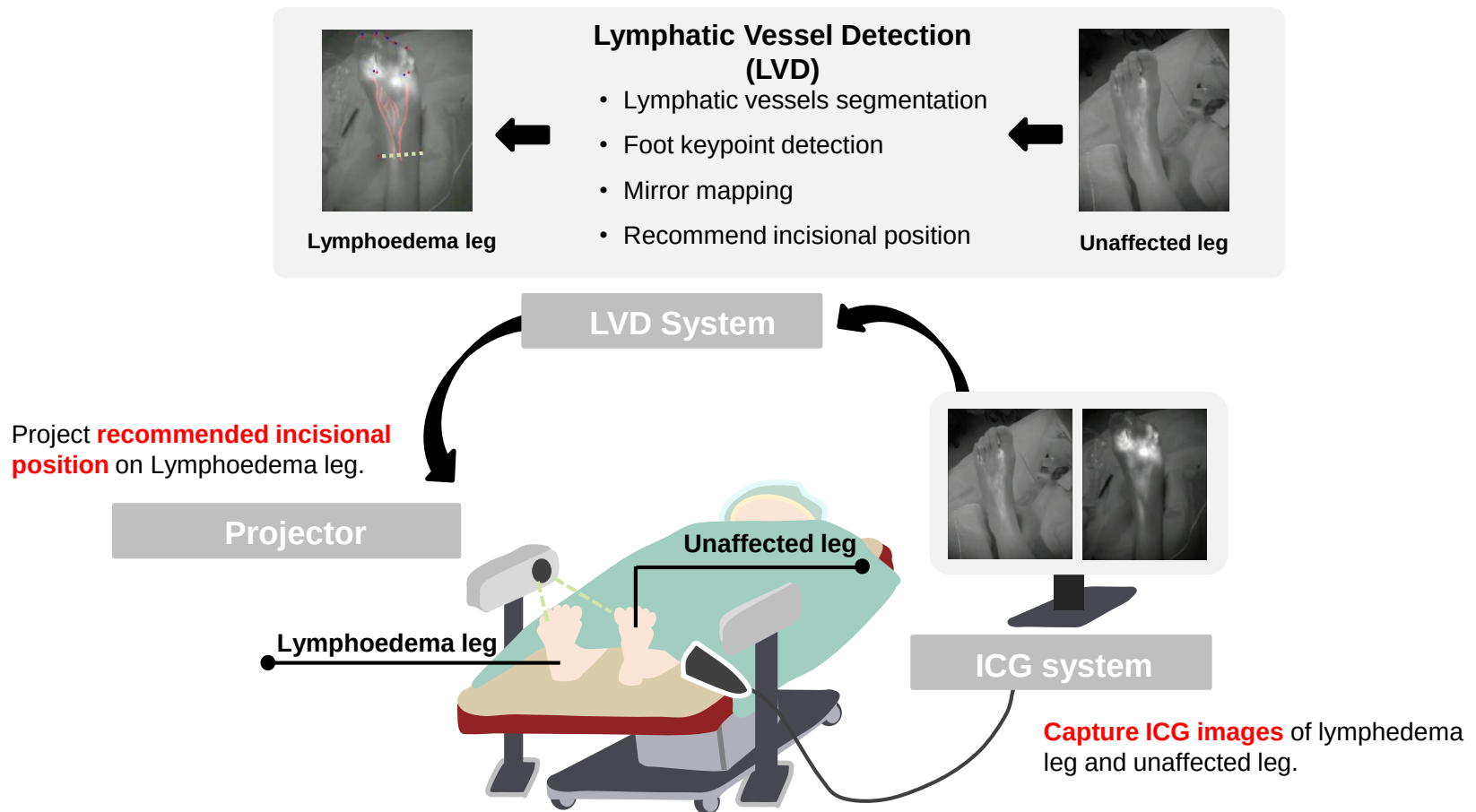
Normal ICG



Edema ICG

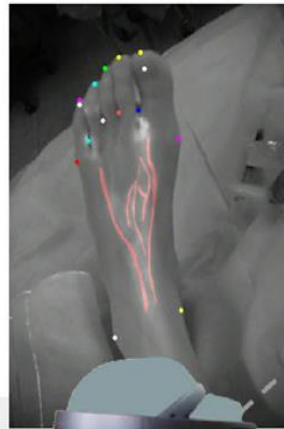


LVA detection model (LVADet)



Lymphatic Detection

Pre-Projection View



Surgeon

Viewpoint #1



Viewpoint #2



Viewpoint #3



Viewpoint #4



Projected View with Hololens



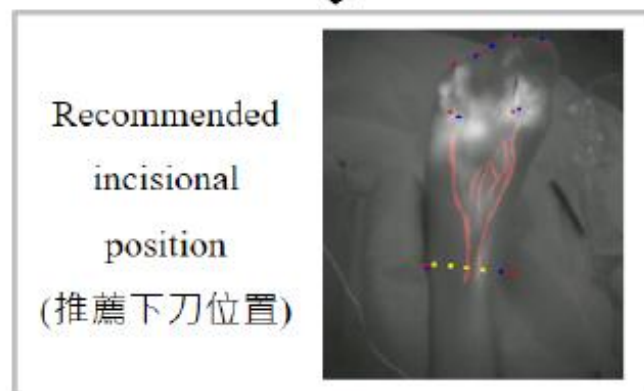
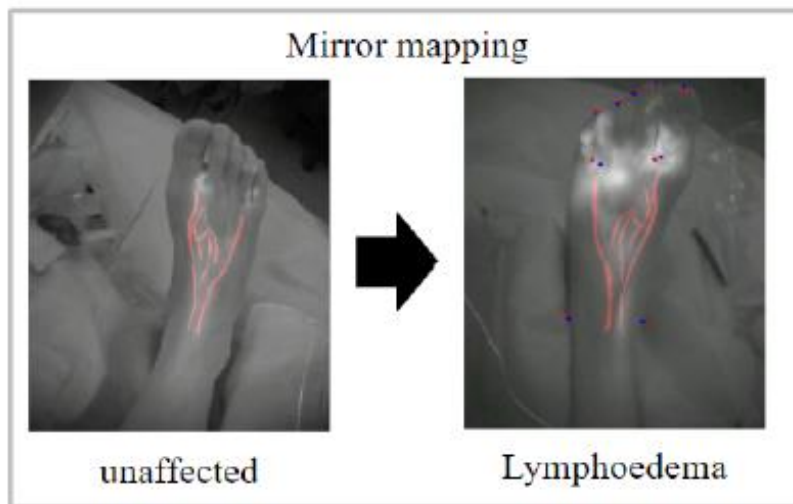
ICG image of unaffected leg



Segmentation

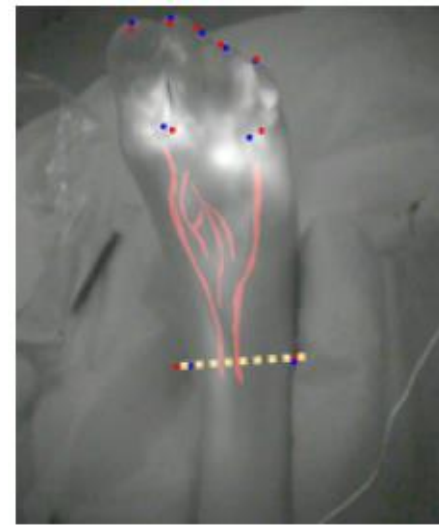


Keypoint detection



淋巴管偵測系統(Lymphatic Vessel Detection) [8]

Conclusion



- **Expected Outcomes of the Project**
 - **Real-time** evaluation of ICG lymphography images using computer vision and machine learning techniques
 - Provision of appropriate **surgical guidance** to improve the overall precision of lymphatic surgery
- **Clinical Applications**
 - **Accurate** localization of lymphatic vessel pathways, allowing smaller surgical incisions
 - **Reduction in operative time** required for lymphatic vessel identification
 - Earlier intervention and optimized treatment strategies may contribute to **reduced healthcare costs**